

# ECN 711: More on Taxes and Debt. Production

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- What is the implication of this?
- Clearly, for a given path of interest rates, individual choices will be the *same* in the two scenarios. Why?

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- The name comes from David Ricardo (1772-1823) who first noted this and after discussing the modes to pay for war, noted:

‘‘In point of economy, there is no real difference in either of the modes ... A man who has 10000 pounds, paying him an income of 500 pounds, out of which he has to pay 100 pounds per annum towards the interest of the debt, is really worth only 8,000 pounds , and would be equally rich, whether he continued to pay 100 pounds per annum or at once, and only once, sacrificed 2,000.’’



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- In the context of our economy, one can prove the following result:

### **Proposition: Ricardian equivalence.**

*Fix an “initial” government (tax and debt) policy and consider an associated “initial” competitive equilibrium. Now consider a new tax policy that preserves the initial policy’s present value (at the initial equilibrium interest rates) of each individual’s tax liability. This policy is equivalent to the initial one in the following sense. There exists a debt policy such that the initial allocation and interest rates, together with the new tax and debt policy, constitute an equilibrium.*

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More involved... induction argument.



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- Preferences: individuals now value streams of consumption ( $c$ ) and **leisure** ( $z$ ).

$$u^h(.) = v^h(c^h(t), z^h(t)) + \beta v^h(c^h(t+1), z^h(t+1)) \quad (1)$$

$v$  is continuous, strictly increasing and strictly concave.  
 $\beta > 0$ .

# Production and labor supply

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- Constraints on time use:

$$1 = z^h(t) + h^h(t); 1 = z^h(t+1) + h^h(t+1)$$

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- Prices for labor services:  $w(t)$  and  $w(t+1)$ .
- Individuals maximize (1), subject to time constraints and lifetime budget constraint:

$$c_t^h(t) + \frac{c_t^h(t+1)}{R(t)} = w(t)h^h(t) + \frac{w(t+1)h^h(t+1)}{R(t)}.$$

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- Profit maximization will imply that  $w(t) = A(t)$ .

# First-order conditions

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- FOCs for individual maximization imply:

$$\begin{aligned}w(t)v_1(t) &= v_2(t) \\w(t+1)v_1(t+1) &= v_2(t+1) \\v_1(t) &= \beta R(t)v_1(t+1)\end{aligned}$$

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- Two *intratemporal* conditions, and one *intertemporal* condition.
- What is the **interpretation** of individual optimality conditions?

# Equilibrium



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- In equilibrium,  $\tilde{w}(t) = A(t)$ , all  $t$ . Likewise,

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- **Exercise:** Define a competitive equilibrium. What objects are involved?
- Note that aggregate feasibility is as usual,  $C(t) = Y(t)$ .
- **Exercise:** show that an equilibrium consumption allocation and an equilibrium allocation of labor are *feasible* if and only if aggregate borrowing and lending is zero at all dates.

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- Identical individuals within a cohort. Preferences:

$$v(c, z) = \log(c) + \frac{z^{1-\sigma}}{1-\sigma}$$

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- $A(t+1)/A(t) = (1+g)$ .
- What is the equilibrium path of interest rates, consumption and hours?



# Example

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- FOC's imply:

$$\frac{w(t)}{c(t)} = (1 - h(t))^{-\sigma}$$

$$\frac{w(t+1)}{c(t+1)} = (1 - h(t+1))^{-\sigma}$$

$$\frac{1}{c(t)} = \beta R(t) \frac{1}{c(t+1)}$$

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- In equilibrium, there is no intertemporal trade. Hence,

$$\tilde{c}(t) = \tilde{w}(t)\tilde{h}(t) = A(t)\tilde{h}(t)$$

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- Hours of work are constant over time.  $\exists \tilde{h} \in (0, 1)$ , unique, that solves:

$$\frac{1}{\tilde{h}} = (1 - \tilde{h})^{-\sigma}$$

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- Why does the growth rate of output affects the interest rate?